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Lesson 6: k-NN, Naive Bayes and Support Vector Machines

Technologies for Artificial Intelligence

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Before we start

- Exercise 4: which threshold did you choose for the classifier, and how did you defend it?
- Exercise 5, set at the end of lesson 5, is discussed next Friday, 20 November

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Today: the first lesson that offers a choice

- **k-nearest neighbours (k-NN)**, and the **curse of dimensionality** that limits it
- **Naive Bayes**: a probability model, very cheap to train
- **Support vector machines (SVM)**: the widest margin, then kernels
- How to choose a model family, and why that matters more than tuning

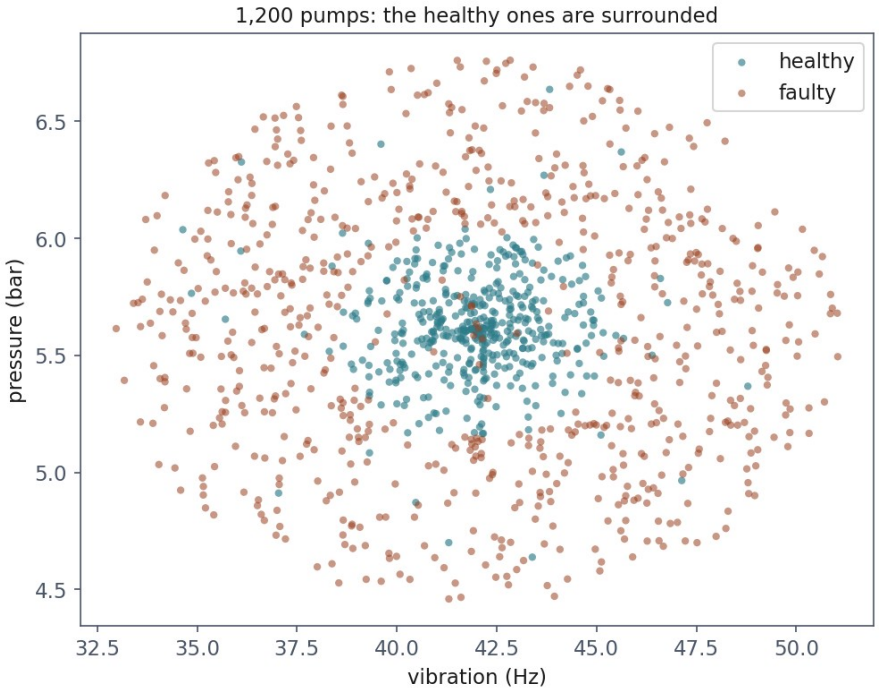
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1,200 pumps, two readings each

- 1,200 simulated pumps, two sensor readings each: vibration in hertz (Hz), pressure in bar
- A pump is faulty when its two readings, taken together, are too far from the **design point, 42 Hz and 5.6 bar**: too low as readily as too high
- 735 of the 1,200 are faulty: **61.3%**
- 4% of the labels are flipped on purpose, as sensor errors: no model can score above about **0.96** (the noise ceiling)

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The shape of the problem: healthy pumps in the middle, faulty all around



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A straight boundary calls all 1,200 pumps faulty

Model	Accuracy, 5-fold cross-validation (mean ± standard deviation over the folds)
Always predict the majority class: 735 ÷ 1,200	0.613
Logistic regression (lesson 4)	**0.613 ± 0.000**

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k-nearest neighbours: the whole algorithm

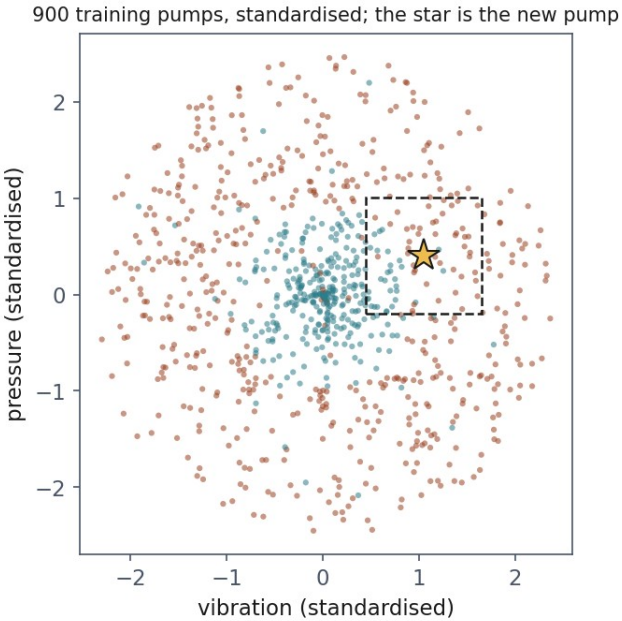
- A line fails, so try a method that assumes no shape: to classify a new pump, take the majority vote of the **k closest** training pumps
- There is no training step: “fitting” only **stores the data** (a **lazy learner**: the work waits until a prediction is asked for)
- In code:

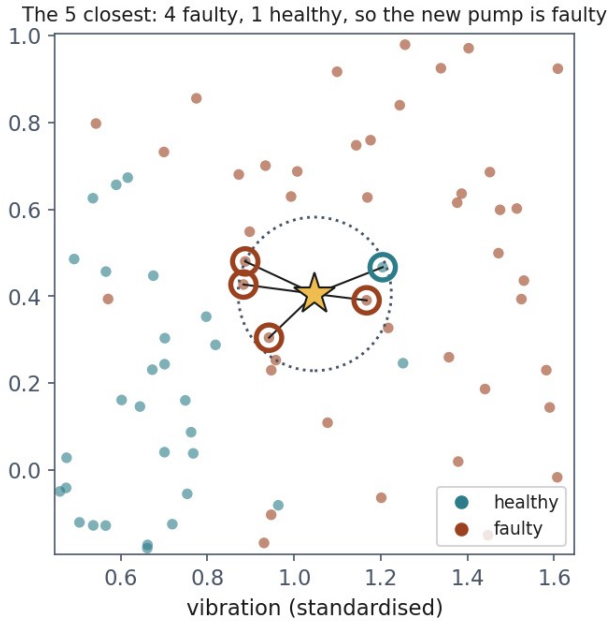
```
KNeighborsClassifier(n_neighbors=5)
```


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k-NN, drawn: one new pump and its five nearest

k-NN with k = 5: 1. standardise 2. measure the distance to every training pump 3. keep the 5 closest 4. majority vote





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Euclidean distance between two pumps x and x′, over their n readings

$$d(x, x') = \sqrt{\sum_{j=1}^{n} (x_j - x'_j)^2}$$

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Scale first, or vibration decides everything

- Vibration spans **18 Hz** (33 to 51), pressure **2.3 bar** (4.5 to 6.8)
- Unscaled, 10 Hz apart is distance 10 and 1 bar apart is distance 1: vibration picks the neighbours
- The fix (lesson 2): **standardise** each column: subtract its mean, divide by its standard deviation
- In code: `make_pipeline(StandardScaler(), KNeighborsClassifier(5))`

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k sets the balance between bias and variance

- **Small k**: the vote follows each stored pump, those with flipped labels included. Low bias, high variance (lesson 5)
- **Large k**: the vote reaches pumps across the edge of the envelope, and near the edge the larger class wins. High bias, low variance
- At k = 1 the training accuracy is **exactly 1.000**: each stored pump is its own nearest neighbour

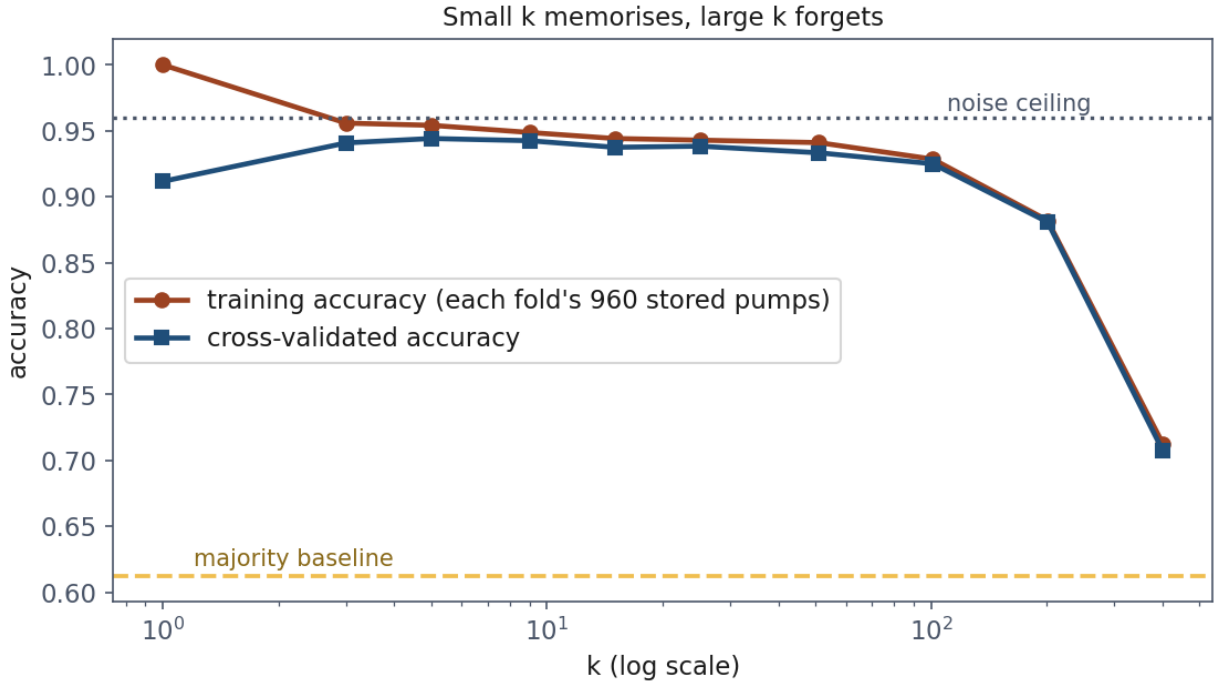
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Training against cross-validated accuracy, by k

k	Training accuracy (on each fold's 960 stored pumps, mean over 5 folds)	Cross-validated accuracy, 5 folds
1	**1.000**	0.912
5	0.954	**0.944**
15	0.944	0.938
51	0.941	0.933
401	0.713	0.708

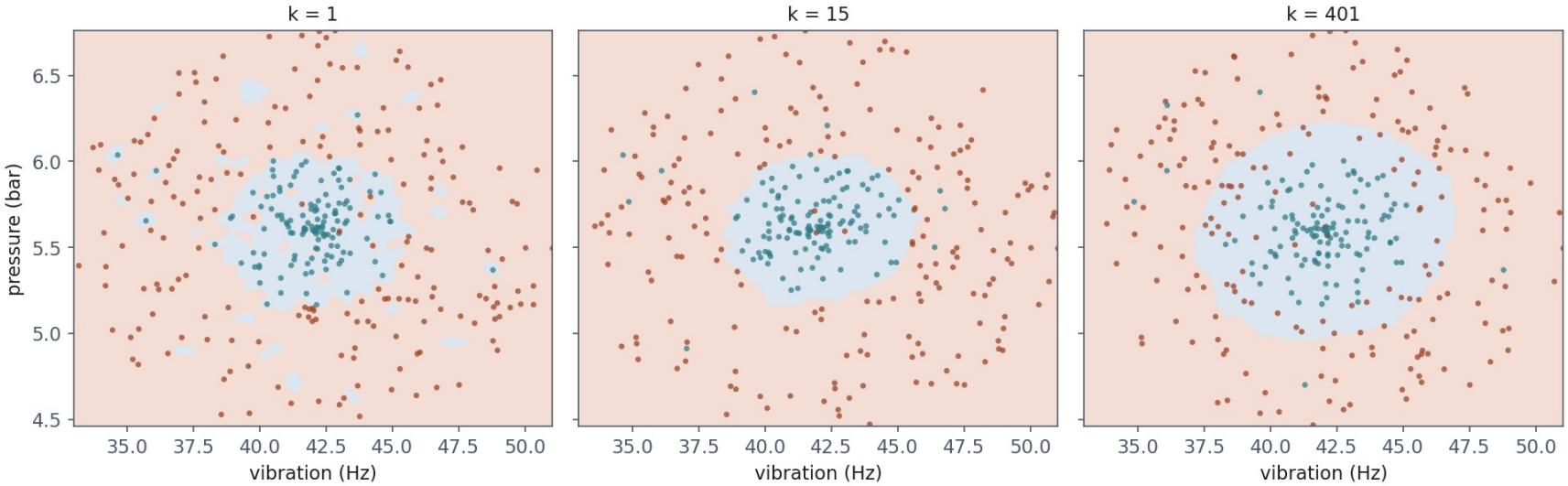
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Accuracy against k, between the baseline and the noise ceiling



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The boundary k-NN draws at k = 1, 15 and 401



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What k-NN costs

- Storing the pumps costs nothing; the cost comes at **every prediction** instead
- A direct implementation compares each new pump with **every** stored pump: about m × n operations for m pumps and n features
- To predict, the model needs every stored pump: **deploying the model means handing over the data**

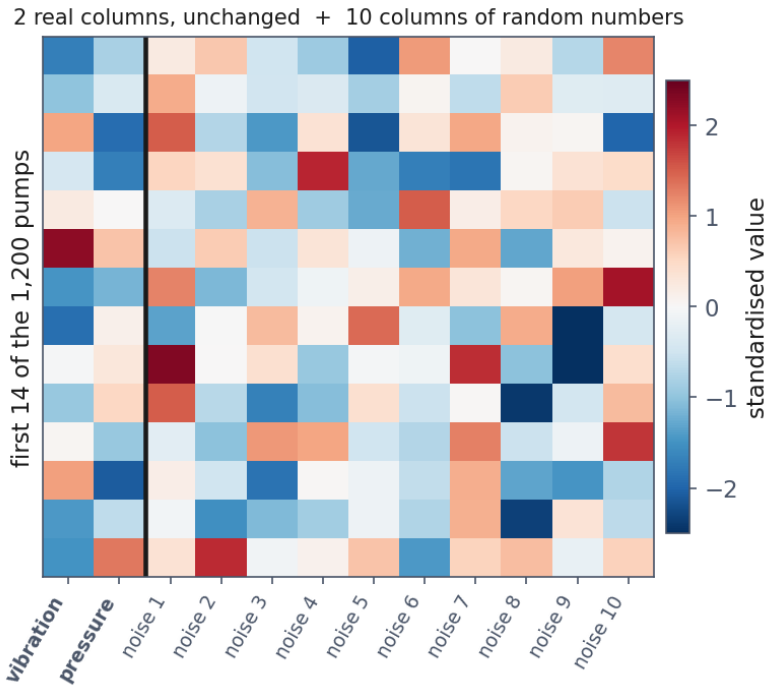
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An experiment: add columns of pure noise

- Keep the 1,200 pumps and their two real readings **unchanged**
- Add columns of **pure noise**: normal random numbers (mean 0, standard deviation 1), unrelated to the label
- Only the number of columns changes: the problem is as solvable as before
- Same model each time: standardised k-NN, k = 15, 5-fold cross-validation

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The experiment, drawn



The same steps for every number of noise columns

1. Keep the 1,200 pumps and their 2 real readings
2. Add columns of random numbers (mean 0, standard deviation 1), unrelated to the label
3. Standardise, then k-NN with k = 15
4. Score it: 5-fold cross-validated accuracy
5. Repeat with 0, 2, 5, 10, 25, 50, 100 noise columns

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The two real readings are still there, and accuracy still falls

Columns: 2 real + noise	Cross-validated accuracy	Accuracy − baseline 0.613
2	0.938	+0.325
12	0.762	+0.149
27	0.662	+0.049
52	0.602	**−0.011**
102	0.578	−0.035

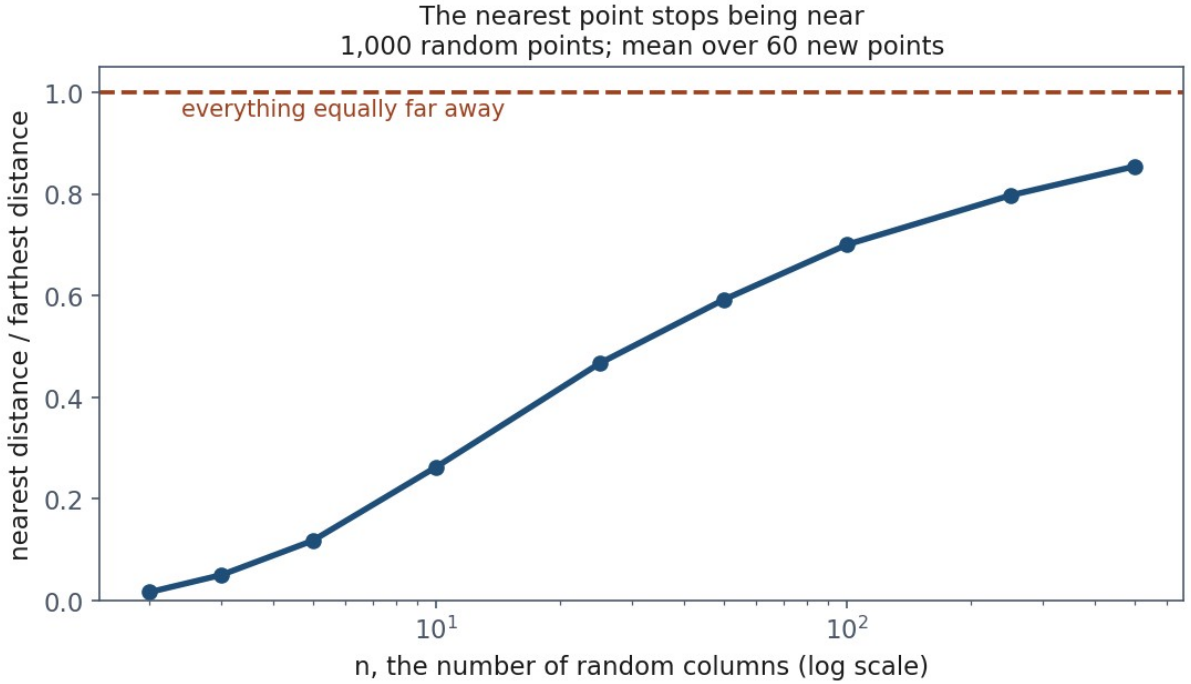
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Why: distances stop varying

- A separate experiment, no pumps: 1,000 random points with n columns, each uniform on [0, 1] (n = 2 a square, n = 3 a cube)
- For a new point, its **nearest** and **farthest** distance: in two dimensions they differ a lot
- Each extra column adds a squared difference to every distance: over many columns **all distances come out about the same**

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With n random columns, the nearest point stops being near



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Nearest divided by farthest

n, random columns	nearest distance ÷ farthest distance, in n dimensions, mean over 60 new points
2	0.016
10	0.263
50	0.592
100	**0.701**
500	0.855

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In 100 dimensions the nearest point is 70% as far as the farthest

- In **two** dimensions the ratio is 0.016: the nearest point is at **2%** of the farthest distance
- In **one hundred** it is 0.701, **70%**: the nearest point is barely nearer than a random one
- Back to the pumps: the 15 nearest are almost 15 pumps at random, so with 52 columns accuracy fell below the baseline

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What the curse is, and is not

- **Not** that many columns cannot be learned: a linear model ignores a useless one
- It is that **distance-based methods stop working**: all distances come out alike
- For k-NN: choose the columns first, inside the pipeline
- In code: `SelectKBest(mutual_info_classif, k=2)` before k-NN: back to **0.938**

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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
the straight line, 4-6	`LogisticRegression`, `cross_val_score`, `StratifiedKFold` → 0.613; `cross_val_predict` → all faulty
k-NN, scaled first, 7-10	`make_pipeline(StandardScaler(), KNeighborsClassifier(k))`
choosing k, 11-14	`cross_validate` with training scores → 0.944 at k = 5
columns of noise, 16-18	k = 15 on `load_with_noise(50)` → 0.602
distances stop varying, 19-22	NumPy: nearest ÷ farthest → 0.701 at n = 100
choosing the columns, 23	`SelectKBest`: `f_classif` 0.562, `mutual_info_classif` 0.938

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Break

- Twelve minutes

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Naive Bayes: ask how likely the readings are in each class

- A second family: we want P(y = c | x), the probability of class c **given** the readings x
- Bayes' rule uses how common each class is, and which readings are common in each (healthy near 42 Hz and 5.6 bar, faulty all around)
- The hard part: how often a class shows **this exact combination** of readings - with many readings, almost no pump of that class is near x

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Bayes' rule, applied to a class

$$P(y = c \mid x) = \frac{p(x \mid y = c)\ P(y = c)}{p(x)}$$

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Drop the denominator: pick the class with the largest numerator (arg max)

$$\hat{y} = \arg\max_c p(x \mid y = c)\, P(y = c)$$

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The assumption: independent, given the class

- To make p(x | y = c) estimable, Naive Bayes assumes: **given the class, the features are independent of one another**
- Then the density of all n readings together is the product of n one-dimensional densities, one per feature
- Each one-dimensional density is estimated from the pumps of that class alone. In code:

```
GaussianNB()
```


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The assumption, written: a product of n densities

$$p(x \mid y = c) = \prod_{j=1}^{n} p(x_j \mid y = c)$$

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Gaussian: each factor is a bell curve

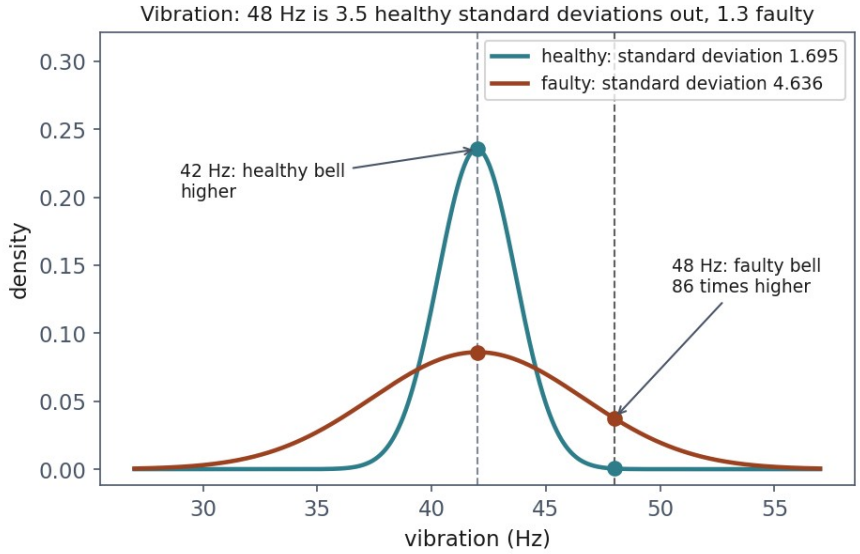
$$p(x_j \mid y = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp\left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2}\right]$$

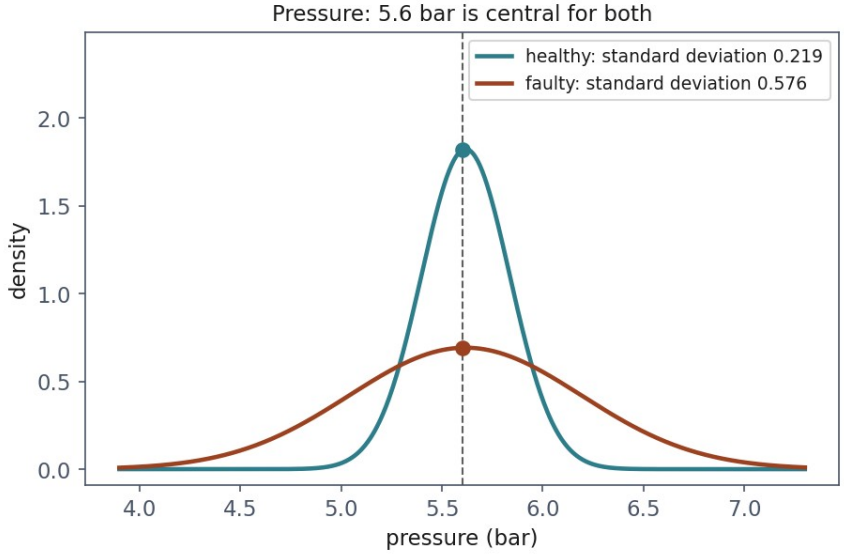
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Training computes ten numbers: the widths separate the classes

Fitted on the 1,200 pumps	Healthy	Faulty
Prior: the class's share of the pumps	465 ÷ 1,200 = 0.3875	735 ÷ 1,200 = 0.6125
Vibration mean	41.986 Hz	42.015 Hz
Vibration standard deviation	**1.695 Hz**	**4.636 Hz**
Pressure mean	5.617 bar	5.616 bar
Pressure standard deviation	**0.219 bar**	**0.576 bar**

A pump at 48 Hz, 5.6 bar: there the faulty bell is 86 times higher





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Logs turn the product into a sum, and keep the same winner

$$\ln\left[P(y=c)\prod_j p(x_j \mid y=c)\right]$$

$$= \ln P(y=c) + \sum_j \ln p(x_j \mid y=c)$$

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48 Hz, 5.6 bar: scores and their logs

Class	Row	Prior	Vibration density	Pressure density	Product of the three, or sum of their logs
Healthy	value	0.3875	0.000433	1.818	0.000305
Healthy	ln	−0.948	−7.745	+0.598	−8.095
Faulty	value	0.6125	0.0374	0.692	0.01585
Faulty	ln	−0.490	−3.286	−0.368	**−4.144**

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From the two totals to probabilities

Step	Healthy	Faulty
1. Log total, previous slide	−8.095	−4.144
2. Undo the log: e^total = prior × densities	0.000305	0.01585
3. Divide each by their sum, 0.000305 + 0.01585 = 0.01616 = p(x)	**0.019**	**0.981**

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Software uses only the gap between the logs

- Many features: e^total rounds to 0 for both classes, and 0 ÷ 0
- The gap stays ordinary: −4.144 − (−8.095) = 3.951
- Divide top and bottom by e^(faulty total): P(faulty) = 1 / (1 + e^(healthy total − faulty total))
- P(faulty) = 1 / (1 + e^(−3.951)) = **0.981**: odds of e^3.951 ≈ **52 to 1**

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A cheap model, built on an assumption that is almost never true

- The gain: training is **one pass** over the data; extra features cost almost nothing
- The assumption rarely holds: on a real pump, vibration and pressure both depend on how hard it runs
- In a text, “New” and “York” appear together far more than independence allows
- So the useful question: **when does being wrong about it cost nothing?**

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Naive Bayes on the pumps: 0.933

Model	Accuracy, 5-fold cross-validation, same folds	Standard deviation over the 5 folds
Majority baseline	0.613	0.000
Logistic regression	0.613	0.000
Gaussian Naive Bayes	**0.933**	0.010
k-NN, k = 5	0.944	0.006

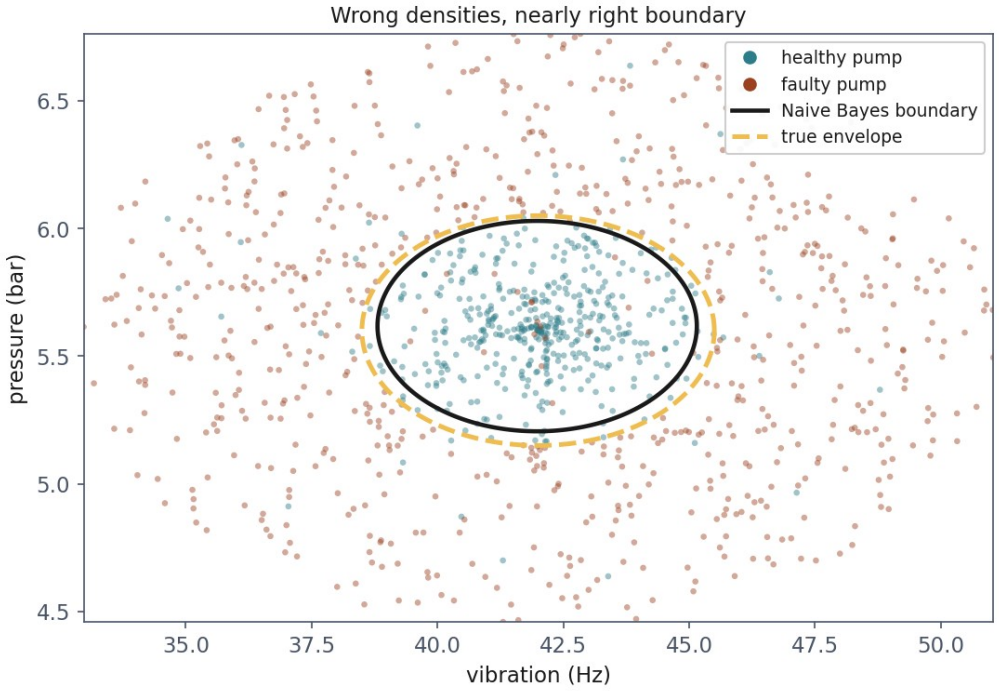
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Uncorrelated within each class, but not independent

Correlation, within one class	between vibration and pressure	between the two standardised readings, squared (how far from the centre)
healthy pumps	−0.006	+0.168
faulty pumps	−0.049	**−0.420**

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The densities are wrong, but the boundary they draw is about right



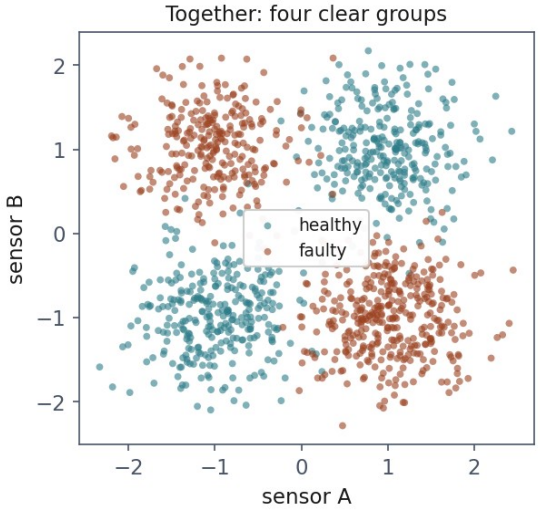
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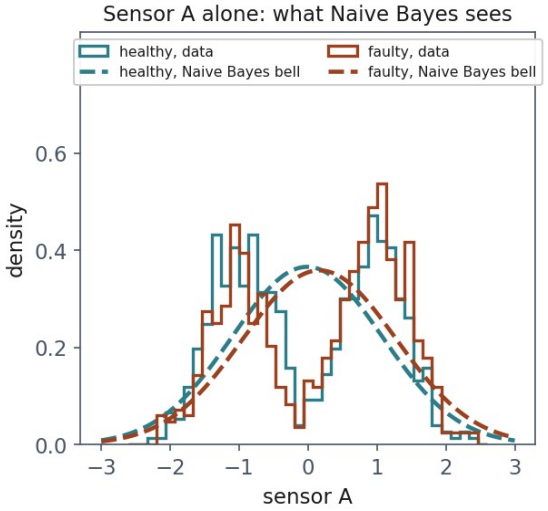
A second dataset: the signal is an interaction

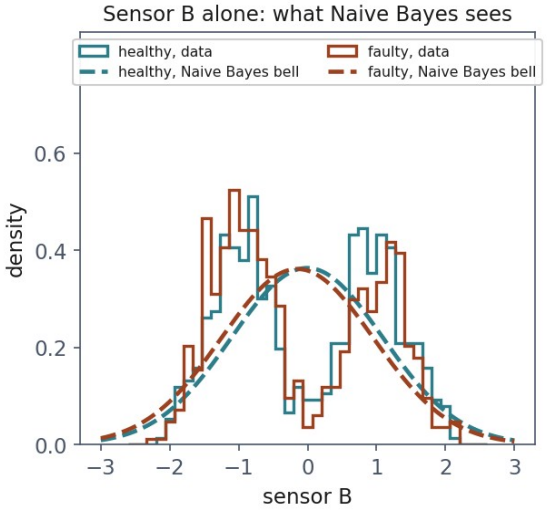
- 1,200 more simulated pumps, with two other sensors, A and B
- Each sensor is either high (around +1) or low (around −1), half the pumps each; faulty when **exactly one** is high - the exclusive or (XOR)
- Both high: high-load mode. Both low: idle. One alone: demand and delivery do not match
- So A alone says nothing: what a high A means depends on B (an **interaction**)

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Each sensor alone: the two classes look the same







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0.404: below the baseline, and below chance

Model	Accuracy, 5-fold cross-validation
Majority baseline: 628 faulty ÷ 1,200	0.523
Gaussian Naive Bayes	**0.404**
Logistic regression	0.393
SVM, linear kernel	0.606
k-NN, k = 5	0.967
SVM, radial basis function (RBF) kernel	0.972

Its probabilities cannot be trusted

- Interacting sensors: Naive Bayes fitted on 900 pumps, tested on the other 300
- For each pump it gives P(faulty) and predicts the larger class. Confidence = the probability of the class it predicts, from 0.5 (a coin) to 1 (certain)
- Mean confidence **0.567** when right, **0.555** when wrong: it cannot tell when it is wrong
- It says about 56% sure and is right on 133 of 300, 44%: its **probabilities are not calibrated**

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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
on the pumps, 39	`GaussianNB` beside `KNeighborsClassifier` and `LogisticRegression` → 0.933
what training computed, 31-37	`.theta_`, `.var_`, `.class_prior_`, `predict_proba` → 0.981 at 48 Hz
checking the assumption, 40-41	`X[y == c].corr()`, within each class → −0.420
the interacting sensors, 42-44	the same models, plus `SVC` → 0.404
confidence, right and wrong, 45	`train_test_split`, then `predict_proba` → 0.567 against 0.555

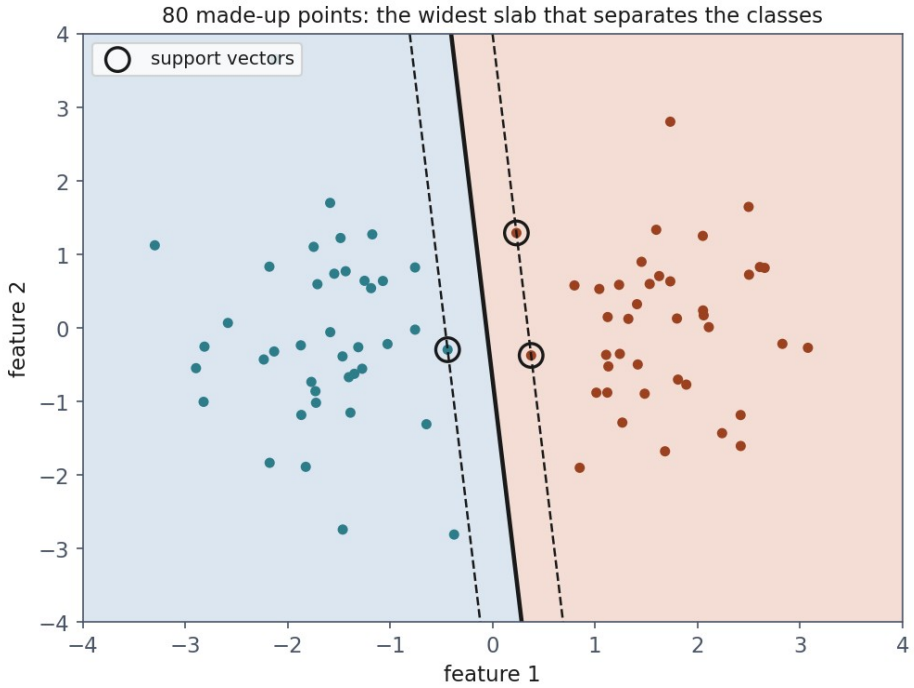
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SVM: among the lines that fit, take the widest margin

- When a line can separate two classes, **infinitely many** lines do, all with **zero training error**: the error cannot choose
- The SVM picks the line with the widest **margin**: an empty strip (the slab) around the line, widened until it touches a point of each class
- The reason: a line that passes close to a point gets it wrong after a small change in that point's readings

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80 made-up points: three of them fix the line (support vectors)



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The strict problem: the widest slab with no point inside

- Labels -1 (healthy), $+1$ (faulty): $y \cdot f(x) > 0$ means “classified right”; in code y stays 0/1
- Boundary $w \cdot x + b = 0$, scaled so the closest points have $|w \cdot x + b| = 1$: the slab is $2/\|w\|$ wide
- Widest slab = shortest w: minimise $\frac{1}{2}\|w\|^2$, every point outside, $y_i(w \cdot x_i + b) \geq 1$
- No term for errors: the constraints forbid them

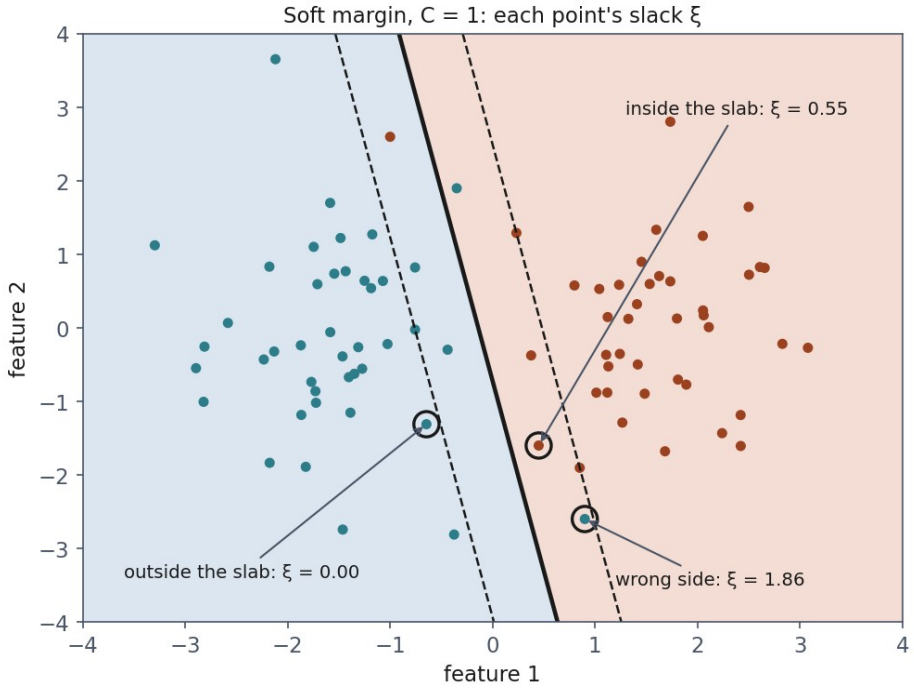
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Not separable? Charge for the violations

- With 4% of the labels flipped, no slab separates the pumps: the strict problem has **no solution**
- Give each point i an allowance $\xi_i \geq 0$, its slack: how far it may fall short of its edge of the slab. Each unit of slack costs C
- $\xi_i = 0$ outside the slab, 0 to 1 inside it, above 1 on the wrong side
- That charge is the **hinge loss** $\max(0, 1 - y \cdot f(x))$, labels ± 1, $f(x) = w \cdot x + b$

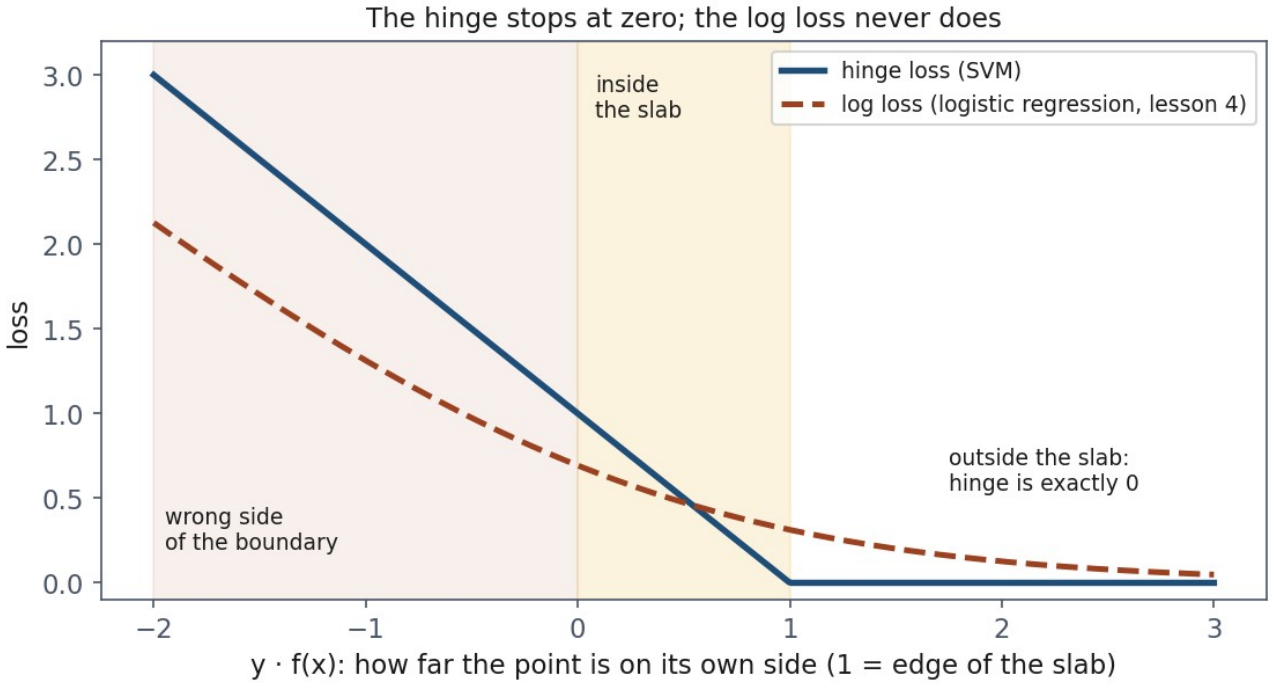
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Three kinds of point: outside the slab, inside it, on the wrong side



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The hinge loss stops at zero



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The SVM objective: small w, plus C times the slack

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

subject to $y_i(w^\top x_i + b) \geq 1 - \xi_i,\ \xi_i \geq 0$

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C is the price of a training error

- **Large C**: slack is expensive, the slab narrows to classify more training points. Low bias, high variance
- **Small C**: slack is cheap, the slab stays wide, some training points stay wrong. Higher bias, lower variance
- C plays the role of k in k-NN and of λ in lesson 3, but **large C means less regularisation**: divided by C, the penalty weighs 1/(2C)

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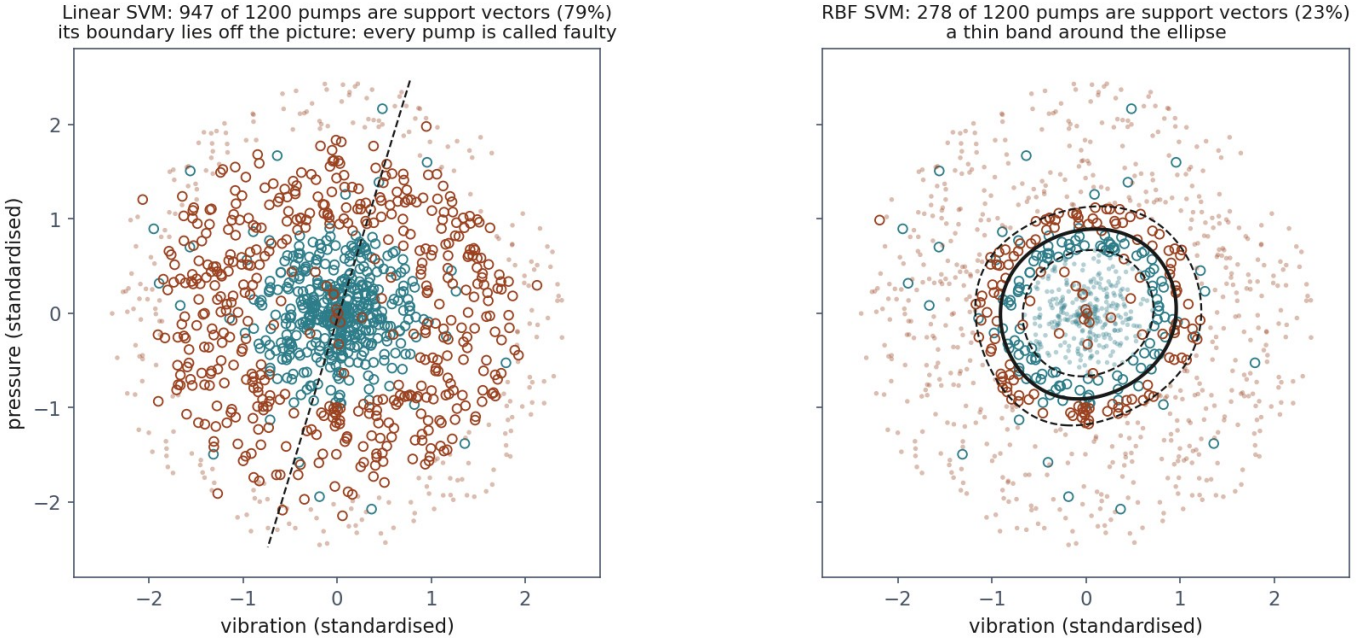
The best straight line is still a straight line

Model, on the pumps (standardised, default C = 1)	Accuracy, 5-fold cross-validation	Support vectors, fitted on all 1,200
SVM, linear kernel: a straight line	**0.613 ± 0.000**	947 of 1,200 (**79%**)
SVM, RBF kernel: a curved boundary	0.947 ± 0.005	278 of 1,200 (23%)

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On the pumps: a straight line leaves almost every pump in its slab

Circled: support vectors, on the edge of the slab, inside it, or on the wrong side. Solid: boundary; dashed: edges of the slab



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Many support vectors: a free warning

- The linear SVM, `SVC(kernel="linear")`, keeps **79%** of the pumps as support vectors; the RBF SVM keeps **23%**
- 79% means almost every pump sits on or inside the slab: no strip of empty space separates the classes
- The count comes free after `fit`, as `len(model.support_)`: a high share is a warning before any cross-validation

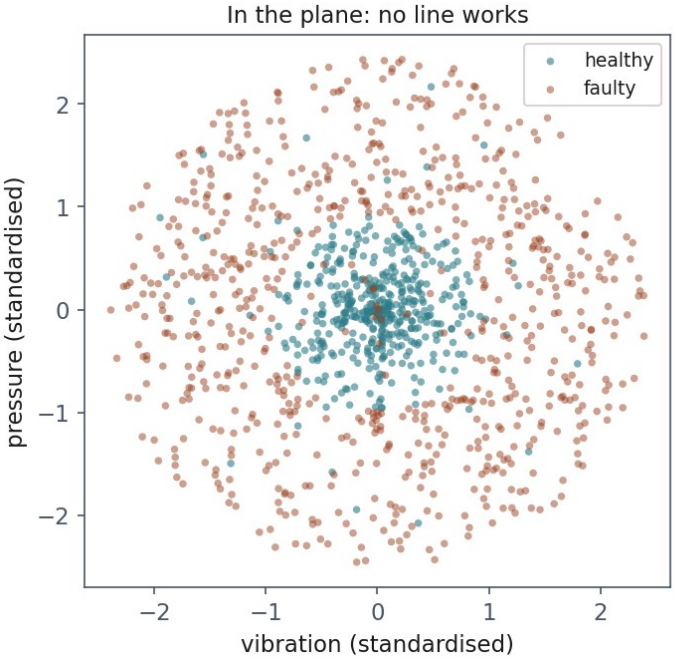
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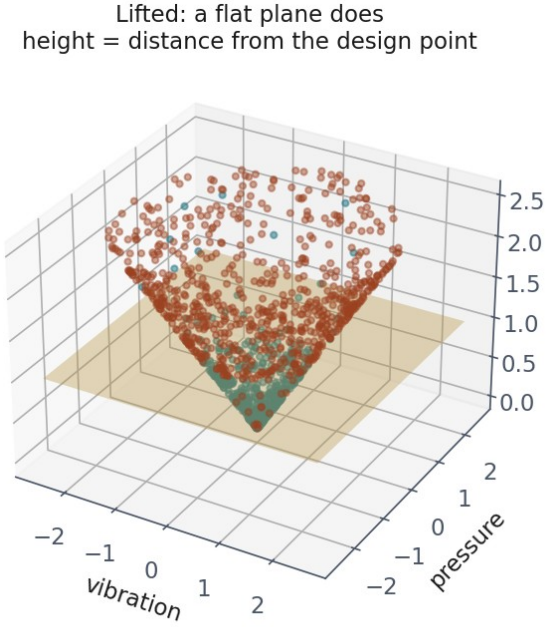
A new coordinate makes the classes separable

- In the plane, healthy pumps sit in a disc, faulty ones around it: no line separates them
- Add a third coordinate: each pump’s **distance from the design point**, standardised
- Healthy pumps rise a little, faulty ones a lot: a flat plane separates them, flipped labels apart
- The classes were separable all along, in other coordinates

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The same pumps, lifted





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The SVM’s solution: a weighted sum of pumps

- Slide 53’s objective: $\frac{1}{2}\|w\|^2 + C \Sigma \max(0, 1 - y_i(w \cdot x_i + b))$
- Its slope in w: $w - C \Sigma y_i x_i$, over the pumps inside the slab or on the wrong side
- Zero slope: $w = \Sigma a_i y_i x_i$; $a_i = C$ inside, 0 outside, 0 to C on the edge
- Only the support vectors have $a_i > 0$: the pegs of the tent

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So the new coordinates are never computed

- $\varphi(x)$: a pump's new coordinates. Slide 60 there: $w = \Sigma\, a_i\, y_i\, \varphi(x_i)$
- Boundary $\Sigma\, a_i\, y_i\, \varphi(x_i)\cdot\varphi(x) + b$; training needs only $\varphi(x_i)\cdot\varphi(x_j)$: the pumps enter **only through inner products**
- A **kernel** $K(x, x')$ gives $\varphi(x)\cdot\varphi(x')$ from the original readings: φ is never computed
- RBF, scikit-learn's default: one exponential per pair

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The kernel trick by hand: 25 = 25

Route, for u = (1, 2) and v = (3, 1)	Computation	Result
Lift with $\varphi(x) = (x_1^2, \sqrt{2}\cdot x_1 x_2, x_2^2)$, then multiply	$\varphi(u) \cdot \varphi(v) = (1, 2.83, 4) \cdot (9, 4.24, 1) = 9 + 12 + 4$	**25**
Never lift: the kernel $(u \cdot v)^2$	$(1\times3 + 2\times1)^2 = 5^2$	**25**

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The RBF kernel: similarity falls with distance

- Radial basis function (RBF) kernel: $K(x, x') = \exp(-\gamma\|x - x'\|^2)$
- A similarity: 1 for identical pumps, towards 0 far apart. With $\gamma = 0.5$: 0.61 at one standard deviation, 0.14 at two
- Why it: expand the exponential and it holds $(x \cdot x')^2$, $(x \cdot x')^3$, … every degree at once - infinitely many new coordinates, no lift to guess
- Cost: one exponential per pair of pumps

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Training: one weight per pump

- fit minimises slide 53’s objective in the weights a_i, two at a time: no learning rate, one answer (RBF, $\gamma = 0.5$, $C = 1$)
- **922** pumps get weight 0: outside the slab, they drop out
- **268** get $C = 1$ (inside the slab or wrong side), **10** sit on its edge: 278 support vectors
- The model kept: those 278 pumps, their weights, and $b = +1.105$

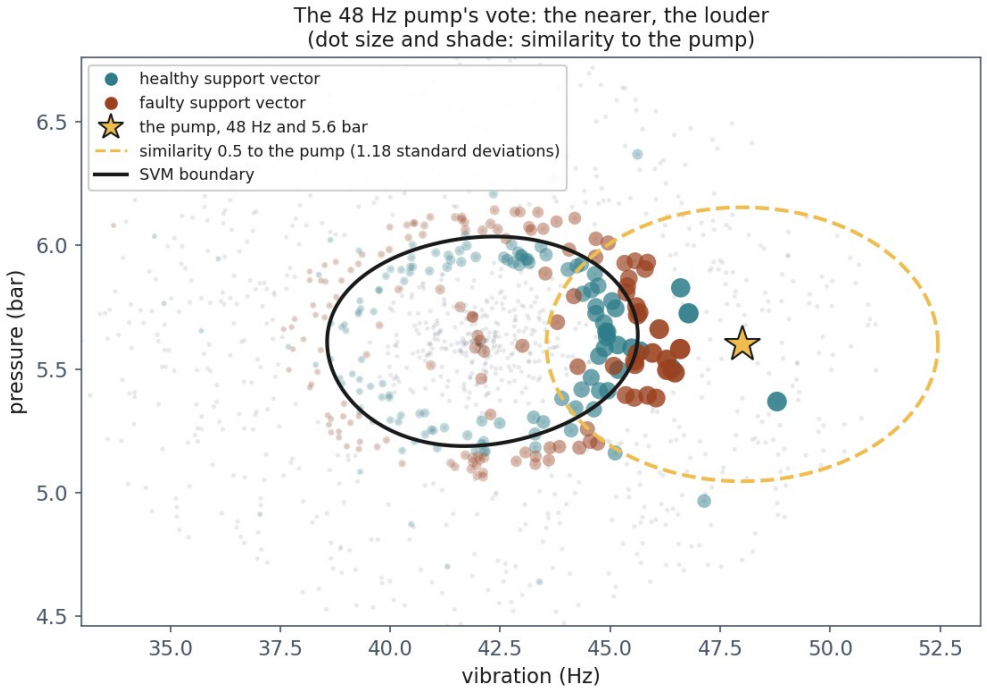
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The SVM’s score at 48 Hz, 5.6 bar: $\Sigma\, a_i\, y_i\, K(x_i, x) + b = +1.883$

Term, for the pump x at 48 Hz, 5.6 bar	Value
Faulty support vectors: sum of $a_i \cdot K(x_i, x)$	+37.455
Healthy support vectors: minus the sum of $a_i \cdot K(x_i, x)$	−36.677
Intercept b	+1.105
Total: positive, so faulty	**+1.883**

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One pump’s vote, drawn



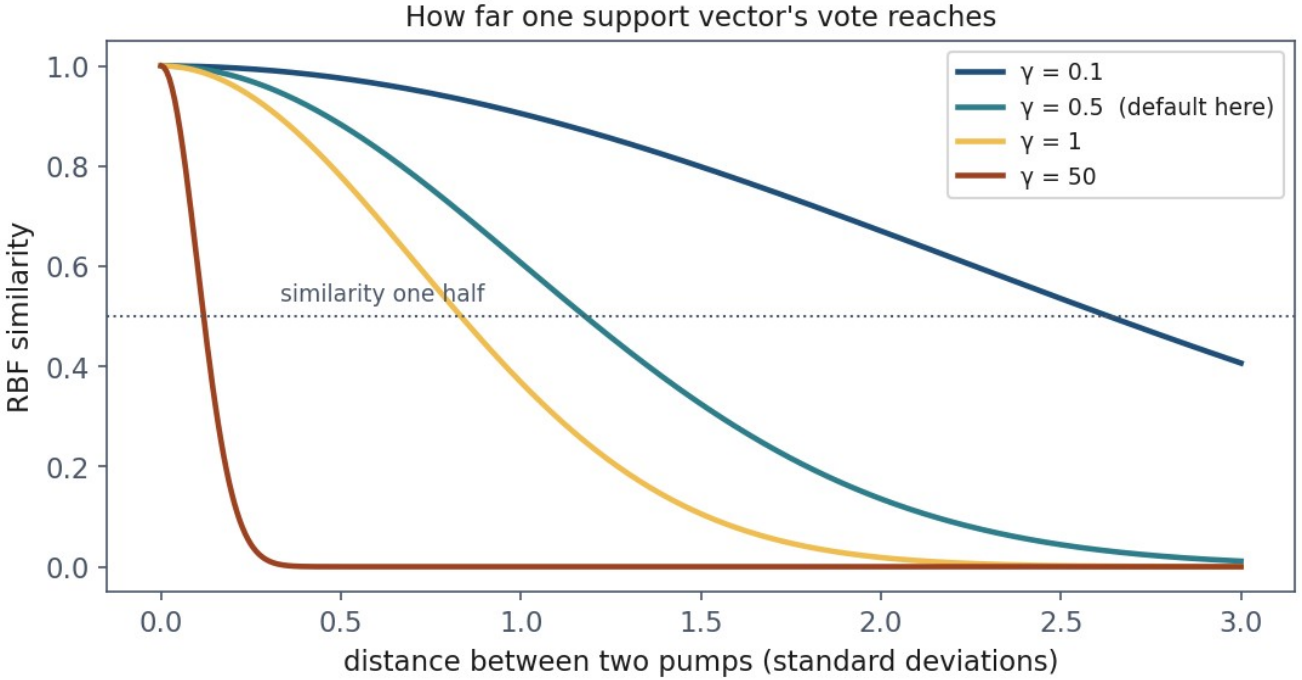
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Training against cross-validated, three settings

RBF SVM on the pumps	Training accuracy (each fold's 960 pumps)	Cross-validated accuracy, 5 folds
$\gamma = 0.1, C = 1$	0.934	0.929
$\gamma = 1, C = 1$	0.948	**0.944**
$\gamma = 50, C = 1000$	**0.996**	0.902

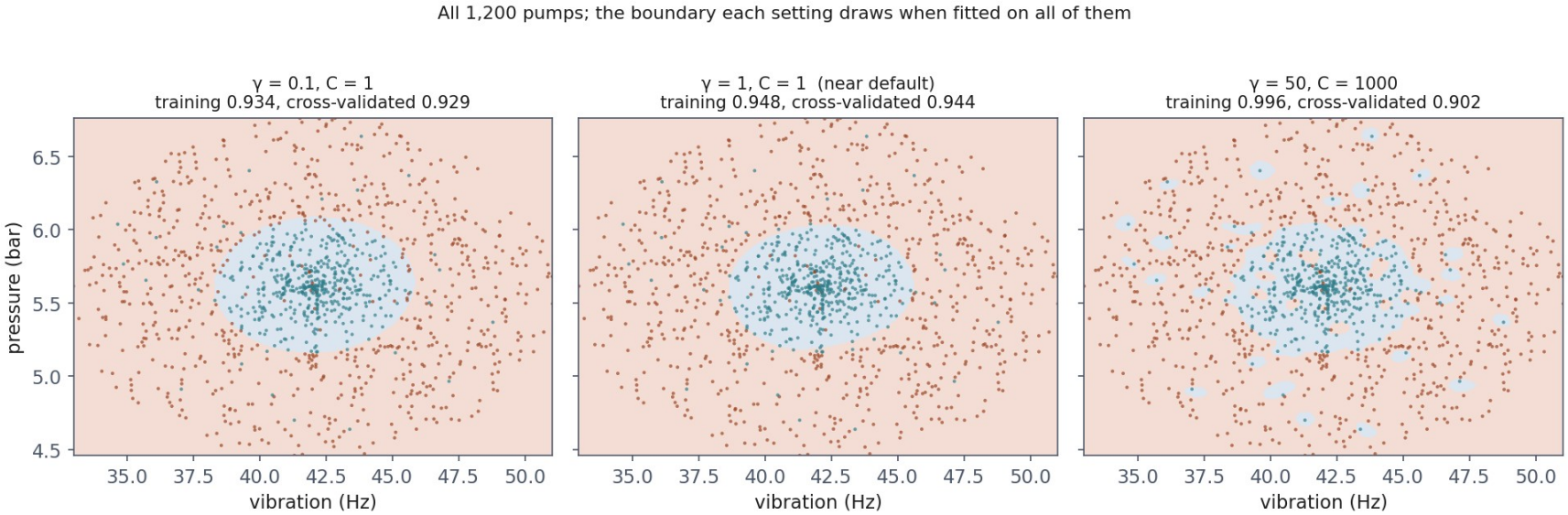
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How far one support vector’s vote reaches, for four values of γ



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Overfitting you can see: the boundary at the three settings



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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the widest slab, 47-52	`SVC(kernel="linear")`, `.n_support_`, `decision_function` → 3 of 80 points
a straight line still fails, 55-57	`cross_val_score`, linear against RBF kernel → 947 against 278 support vectors
the kernel trick, 58-63	NumPy only: $(u \cdot v)^2$ against the lifted product → 25 = 25
training and one prediction, 64-66	`.support_`, `.dual_coef_`, `.intercept_` → +1.883
γ and C, 67-69	`SVC(gamma=..., C=...)`, `cross_validate` → 0.944 at γ = 1

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The three families compared, on the same pumps and the same folds

Model	Accuracy, 5-fold cross-validation	Standard deviation over the 5 folds
Majority baseline	0.613	0.000
Logistic regression (lesson 4)	0.613	0.000
SVM, linear kernel	0.613	0.000
Gaussian Naive Bayes	0.933	0.010
k-NN, k = 5	0.944	0.006
SVM, RBF kernel	**0.947**	0.005
noise ceiling	*≈ 0.96*	

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How to choose, in practice

Situation	Reach for
Few features, plenty of data, a curved boundary	k-NN, or an RBF SVM
Very many features, little data	Naive Bayes
A line separates the classes	Logistic regression or a linear SVM
You need probabilities you can trust	Logistic regression (lesson 4); **not** Naive Bayes, and an SVM gives only a score
The model must be small, or fast at prediction	Naive Bayes; an SVM with few support vectors; not k-NN
The signal is an interaction between features	k-NN or an RBF SVM, not Naive Bayes

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What to take away

- **No line separates the pumps**: 0.613
- **k-NN**: k sets bias against variance; in 100 random columns the nearest is 70% as far as the farthest
- **Naive Bayes**: 0.933, but **0.404** on interactions
- **SVM**: support vectors decide; kernels bend the line
- **Choosing the family beats tuning**: 0.613 to 0.947

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Homework

- **Exercise 6**, discussed at the start of **Friday 20 November**
- You get a dataset, fit the three families, and argue which one you would choose, and why

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